

Getting Started with Python for Data Engineering on Databricks

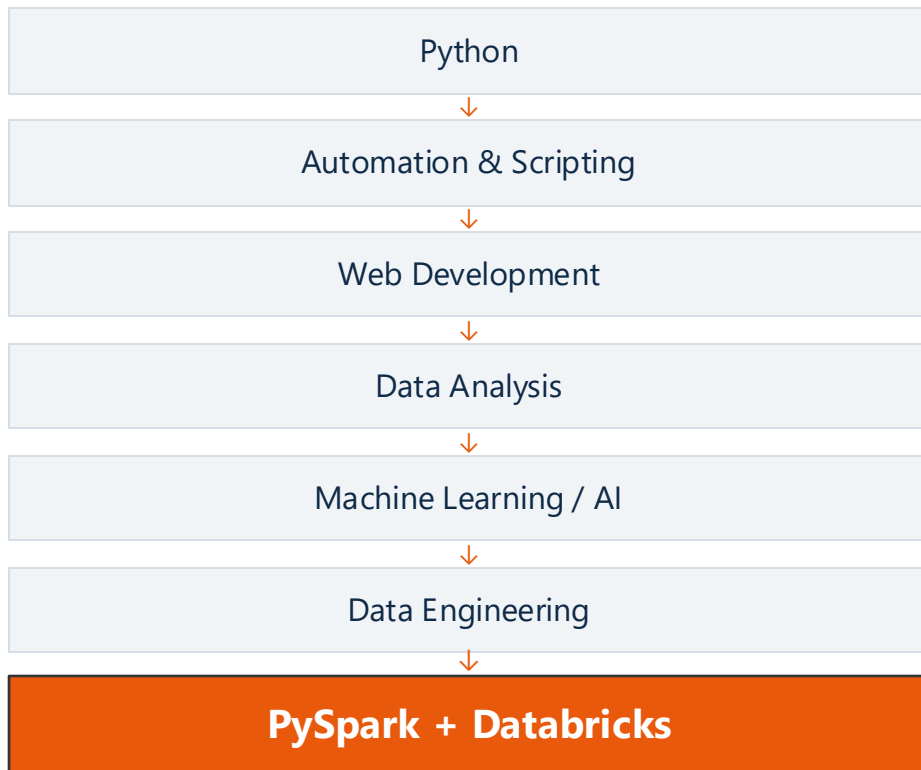
Python → PySpark → Distributed Data → Databricks

THE QUESTION

If Python started as a general-purpose language...

Why is it now one of the most popular languages in Data Engineering?

The Python Journey



Python in the Real World

Web Development

- REST APIs
- Backend services
- Authentication
- Microservices
- Web applications

Automation

- File processing
- Excel automation
- Browser automation
- ETL scripts
- DevOps automation

Data

- Analysis
- Transformation
- Visualization
- Machine Learning

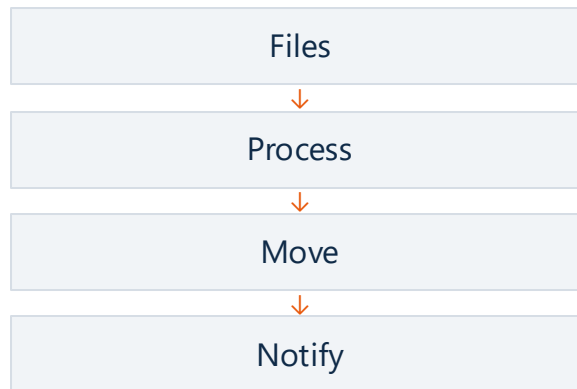
Same language, three very different jobs — and Data Engineering borrows from all of them.

Python for Automation

```
import os
import shutil

for file in os.listdir("input"):
    if file.endswith(".csv"):
        shutil.move(
            f"input/{file}",
            f"processed/{file}"
        )
```

Python can automate repetitive work.



Python for Data Analysis

```
import pandas as pd

df = pd.read_csv("sales.csv")

df.groupby("region")["revenue"].sum()
```

Simple.
Expressive.
Powerful.

But...

What happens when the data no longer fits on one machine?

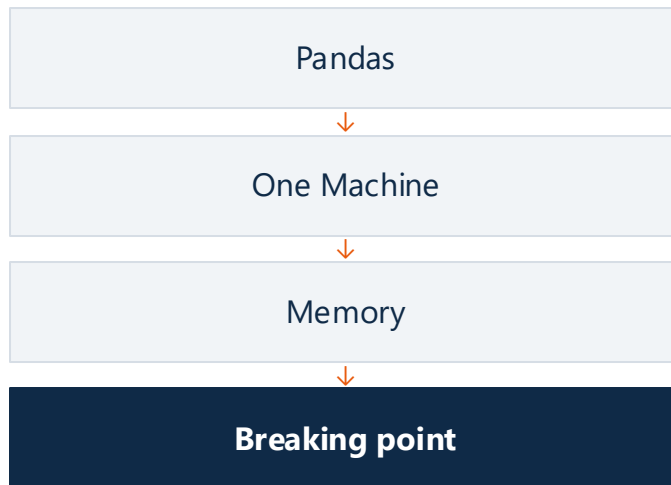
The Big Data Question

What happens when:

sales.csv = 500 MB

becomes

sales.csv = 5 TB?



We need a different execution model.

REASON 1

Why Python for Data Engineering?

A huge, mature ecosystem

Data

Pandas • NumPy • PyArrow • Polars

Data Engineering

PySpark • Airflow • Dask • dbt integrations

Cloud

AWS SDK • Azure SDK • GCP SDK

AI / ML

Scikit-learn • PyTorch • TensorFlow

Python's Productivity Advantage

```
df["salary"].mean()
```

One line. Zero ceremony.

High productivity

+

Rich ecosystem

+

Readable syntax

= High productivity-to-complexity ratio

"Python doesn't have to be the fastest language to be extremely effective."

Python vs Other Languages

Area	Python	Java / Scala	SQL	C++
Learning curve	★★★★★	★★★	★★★★	★★
Data ecosystem	★★★★★	★★★★	★★★★	★★
Development speed	★★★★★	★★★	★★★★	★★
Data analysis	★★★★★	★★★	★★★★	★
Distributed processing	★★★★★	★★★★★	★★★★	★★★★
AI / ML integration	★★★★★	★★★	★★	★★★★
Raw performance	★★★	★★★★★	Depends	★★★★★

Python is often the productivity layer — the engine underneath can still be optimized JVM or native code.

Analyst vs Data Engineer

Data Analyst

Question

"What happened?"

Examples

- Sales by region
- Monthly revenue
- Customer demographics

Tools

SQL • Pandas • Excel • Power BI • Tableau

Data Engineer

Question

"How do we reliably get, transform, store and serve this data?"

Examples

- Ingestion pipelines
- Storage and data models
- Orchestration and monitoring

Tools

Python • SQL • Spark • Cloud • Databases

What Does a Data Engineer Build?

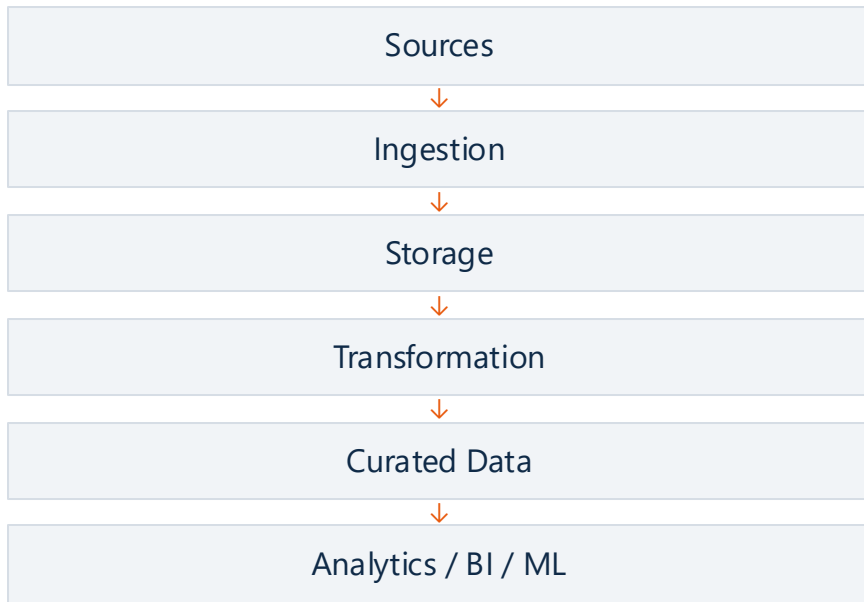
Database

APIs

Files

Events

Logs



Python can participate at almost every stage.

A Real-World Data Engineering Problem

DATA

- Orders
- Customers
- Payments
- Products
- Website events
- Mobile events

SOURCES

MySQL
REST APIs
CSV / JSON
Kafka
Application Logs

Goal: build reliable, scalable data pipelines for analytics and ML.

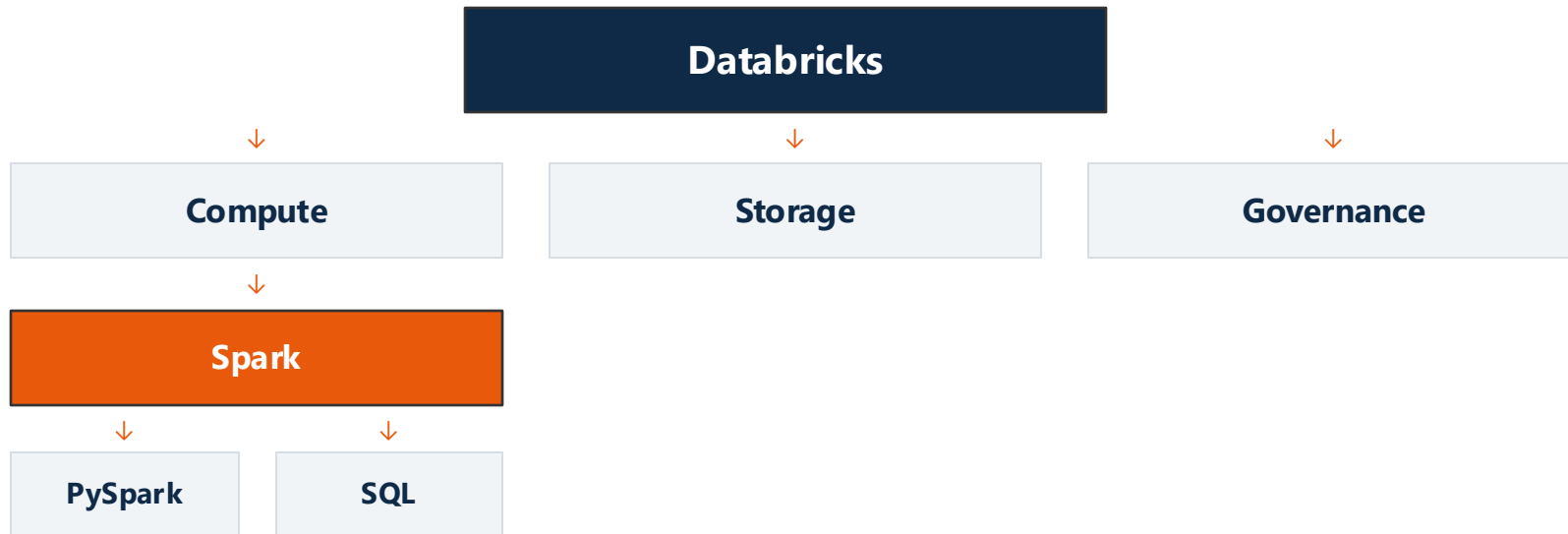
The Data Pipeline



This is where Databricks enters the picture.

What is Databricks?

A platform for building Data + Analytics + AI workflows



Apache Spark is a major processing engine within the platform.

Meet PySpark

```
import pandas as pd

df = pd.read_csv("orders.csv")

result = (
    df.groupby("customer_id")["amount"]
      .sum()
)
```

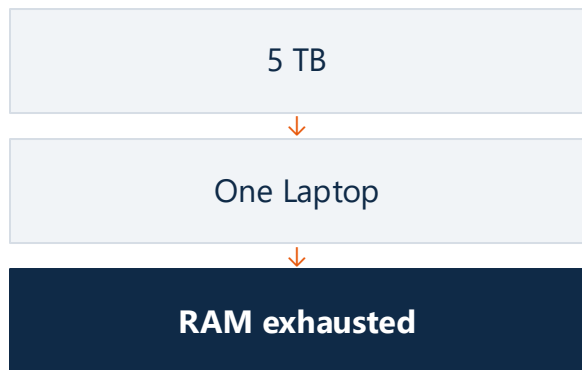
Great when the data can reasonably be processed on one machine.

But what if...

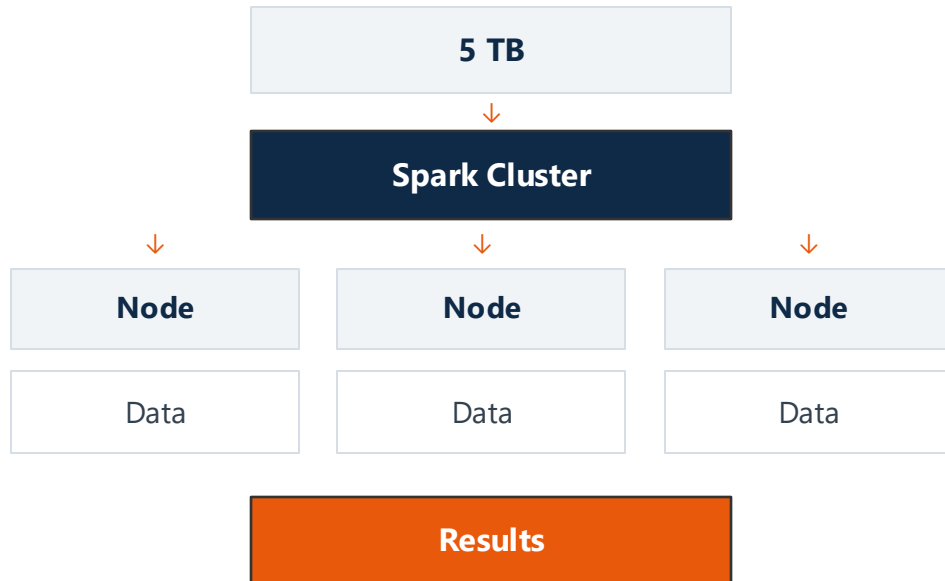
orders.csv = 5 TB?

From One Machine to a Cluster

Pandas



Spark



That's the conceptual reason for Spark.

Pandas → PySpark

PANDAS

```
result = (  
    df.groupby("customer_id")["amount"]  
        .sum()  
)
```

PYSPARK

```
from pyspark.sql import functions as F  
  
result = (  
    df.groupBy("customer_id")  
        .agg(  
            F.sum("amount").alias("total_amount")  
        )  
)
```

Familiar Python syntax



Distributed execution

PySpark Fundamentals

READ

```
df = spark.read.csv(  
    "/data/orders.csv",  
    header=True,  
    inferSchema=True  
)
```

SELECT

```
df.select("customer_id", "amount")
```

FILTER

```
df.filter(df.amount > 1000)
```

GROUP

```
df.groupBy("customer_id").agg(  
    F.sum("amount").alias("total")  
)
```

PySpark: Join & Write

JOIN

```
orders.join(  
    customers,  
    orders.customer_id == customers.id  
)
```

WRITE

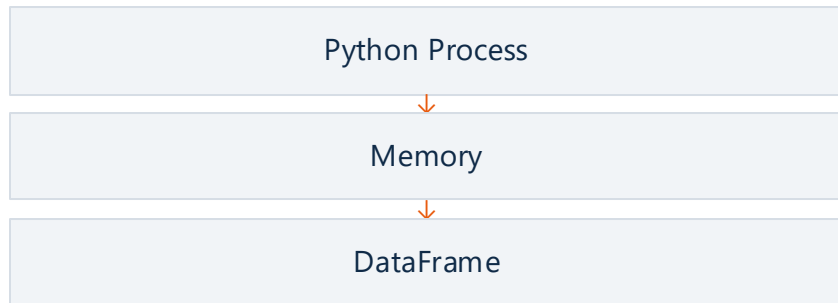
```
df.write \  
    .format("delta") \  
    .mode("overwrite") \  
    .save("/data/orders")
```

The goal isn't to memorize APIs.

Understand the mental model.

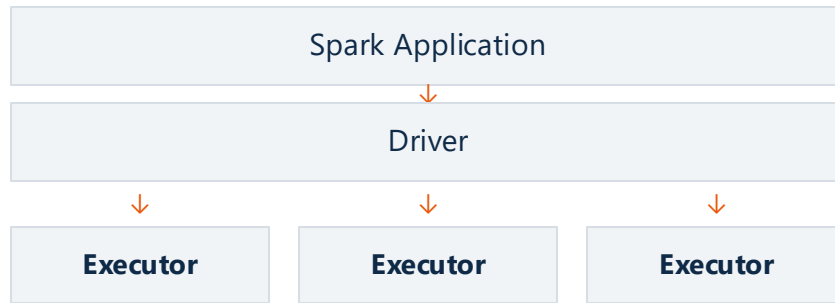
Pandas vs PySpark

PANDAS



Single machine, in-memory processing

PYSPARK



Distributed processing across multiple machines

PySpark isn't simply "Pandas for big data."

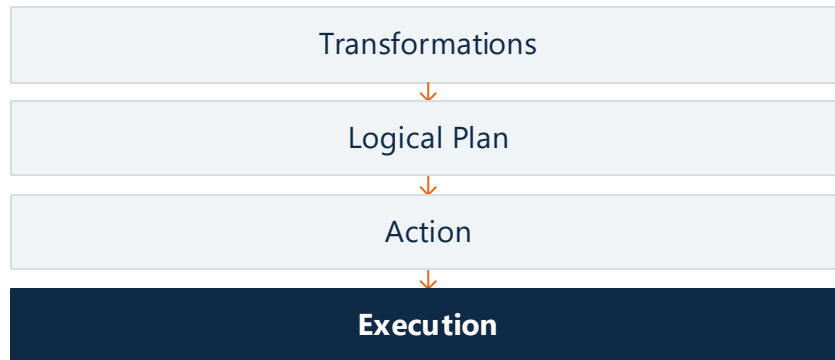
The execution model is fundamentally different.

Lazy Evaluation

```
df2 = df.filter(df.amount > 1000)

df3 = df2.select(
    "customer_id",
    "amount"
)
```

At this point, Spark only builds a plan.



Transformations

`select()` • `filter()` • `join()` • `groupBy()` • `withColumn()`

Actions

`show()` • `count()` • `collect()` • `write()`

Databricks Hands-On Demo

Read Data



Explore



Transform



Aggregate



Write Delta

```
from pyspark.sql import functions as F

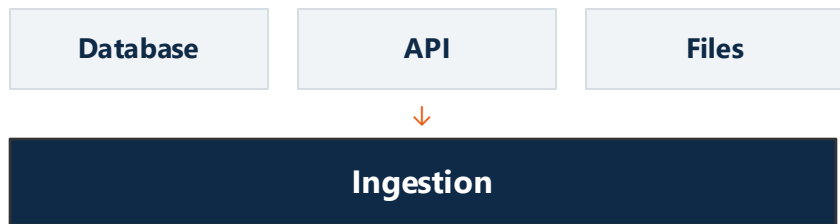
orders = (
    spark.read
    .option("header", True)
    .option("inferSchema", True)
    .csv("/databricks-datasets/...")
)

summary = (
    orders
    .groupBy("country")
    .agg(
        F.sum("amount").alias("revenue"),
        F.count("*").alias("orders")
    )
    .orderBy(F.desc("revenue"))
)
```

Live demo: Explore → Transform → Aggregate → Visualize

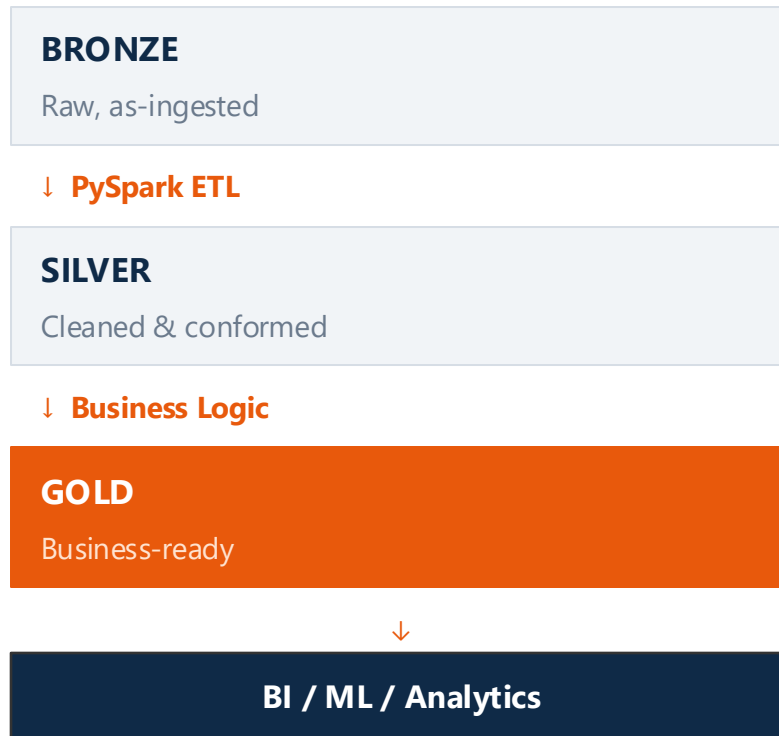
A Real Data Engineering Architecture

DATA SOURCES

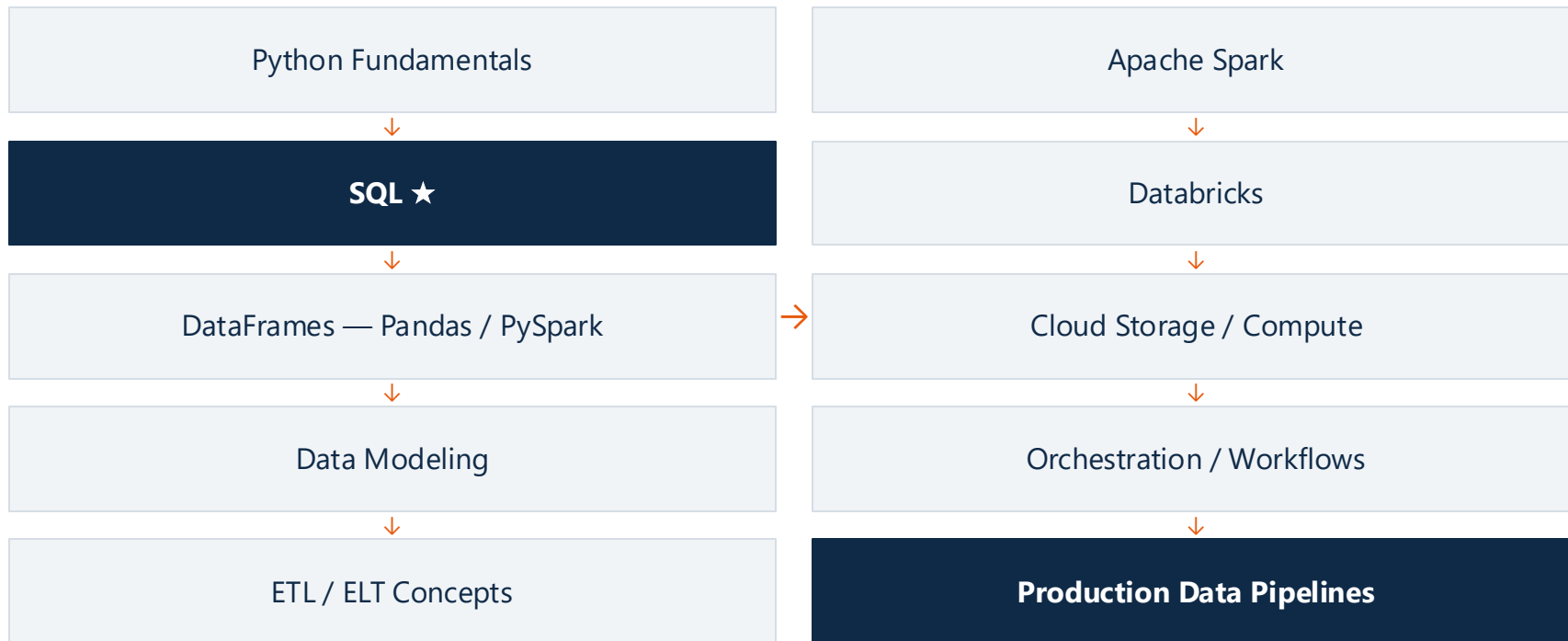


Bronze → Silver → Gold

Each layer raises data quality and trust, so downstream consumers build on curated tables rather than raw feeds.



From Python Developer to Data Engineer



What You Need to Learn

MUST HAVE

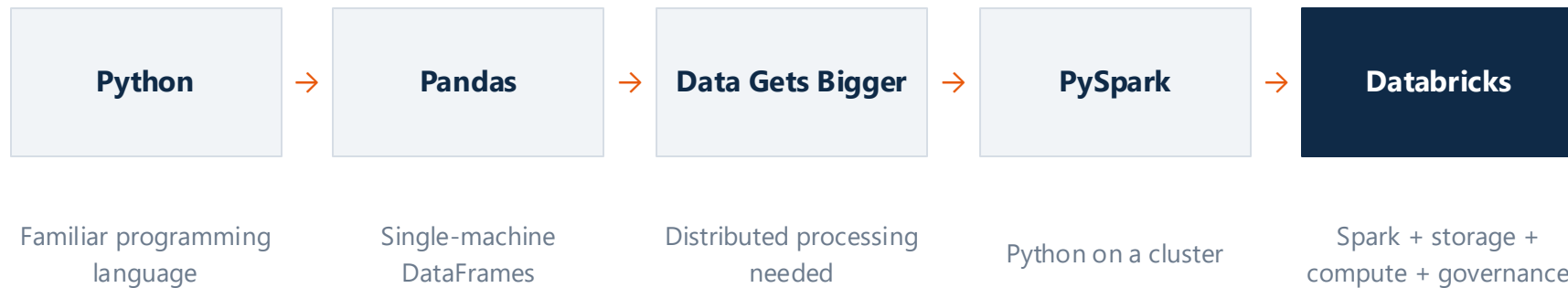
Python	SQL ★	Data structures	ETL / ELT	Data modeling
Spark / PySpark	Cloud fundamentals	Git	Testing	Monitoring

THEN SPECIALIZE

Databricks	AWS / Azure / GCP	Kafka	Airflow	dbt	Delta Lake	Streaming
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Depth in the fundamentals first — tools are much easier to add afterwards.

The One Story to Remember



You already know Python.

Now learn how Python changes when the data becomes distributed.

The 5-Minute Takeaway

1

Python is more than a programming language.

2

Data Engineering needs more than Python.

3

Pandas is excellent — until scale changes the problem.

4

PySpark brings Python to distributed data processing.

5

Databricks brings the platform, compute and workflow together.

You already know Python.

Now learn to think in data.

Python → PySpark → Spark → Databricks

Questions?